Q1

2.1

Let "kids" denote the number of children ever born to a woman, and let "educ" denote years of education for the woman. A simple model relating fertility to years of education is

$$kids = \beta_0 + \beta_1 educ + u,$$

where u is the unobserved error.

(i)

What kinds of factors are contained in u? Are these likely to be correlated with level of education?

Education of the woman's parents, number of siblings the woman has, her income, her marital status, her job, etc., are all likely to be contained in u. That is, each of these factors could be thought of as "explanatory." Each of them (and many others that you could come up with) help determine the number of children a woman has. Many of the factors that I just listed are likely to be correlated with a woman's level of education. Education of the woman's parents: certainly correlated. Number of siblings the woman has: arguable whether or not this is correlated with the woman's level of education. Marital status: correlated. Job: correlated.

(ii)

Will a simple regression analysis uncover the ceteris paribus effect of education on fertility? Explain.

Because the omitted explanatory variables (such as those above) are embodied in u, and because these variables are also correlated with education, the regression will not reveal a causal relationship. The assumption $E(u|x) = 0$ is violated.

Q2

2.4

The data set BWGHT.RAW contains data on births to women in the United States. Two variables of interest are the dependent variable, infant birth weight in ounces (bwght), and an explanatory variable, average number of cigarettes the mother smoked per day during pregnancy (cigs). The following simple regression was estimated using data on n = 1,388 births:

$$\widehat{bwght} = 119.77 - 0.514cigs$$

(i)

What is the predicted birth weight when cigs = 0? What about when cigs = 20 (one pack per day)? Comment on the difference.

The predicted birth weight when cigs = 0 is 119.77 ounces (7.5 lbs). When cigs = 20, the predicted birth weight is equal to 119.77 - 0.514 * 20 = 109.5 ounces (6.8 lbs). It seems clear to me from this equation that (shock!) smoking is nasty bad: it makes your breath stink, your teeth yellow, makes it hard to run a mile, and it makes your kid small (OK, only the last one is justified by the model, but you know what I mean).

(ii)

Does this simple regression necessarily capture a causal relationship between the child's birth weight and the mother's smoking habits? Explain.

The regression captures a causal relationship between the child's birth weight and the mother's smoking habits *unless* the mother's smoking habits are positively related to other unobserved factors which also negatively influence birth weight. Off the top of my head, I would guess that income and education (which both influence nutrition and the amount of prenatal care that a woman gets) could be such variables. So if we wanted to be sure that smoking reduced birth weight, we'd have to make sure that these variables were accounted for. My guess is that we would still find a negative relationship between smoking and birth weight, but that the effect would be attenuated (i.e. the coefficient would be smaller in absolute value).

(iii)

To predict a birth weight of 125 ounces, what would cigs have to be? Comment.

Uh-oh. Gotta drop some algebra to get this one done. 125 is the y-variable, in this case.

$$125 = 119.77 - 0.514 * cigs.$$

Solve for cigs:

$$cigs = \frac{119.77 - 125}{0.514} = -10.2.$$

Impossible! You can't smoke negative cigarettes (if anybody figures out how to do this please let me know. I'd like to begin negative smoking immediately).

(iv)

The proportion of women in the sample who do not smoke while pregnant is about 0.85. Does this help reconcile your finding from part (iii)?

I don't really like this question. It's not as illuminating as you might think. As I mentioned in class, this *sort of* helps reconcile the findings. If most of our sample is pegged at cigs = 0, then I expect that the intercept is going to be pretty close to the mean birth weight for the part of the sample that doesn't smoke (think of the interpretation of the intercept and you'll see why). So even though some children a certainly born with birth weights above 120, our model doesn't say much about them. All of the stuff that causes children to be born heavier than about 120 ounces is unobservable in our model.